INDIAN INSTITUTE OF TECHNOLOGY KANPUR

SPA617 — SPASE INTRUMENTATION LABORATORY

ANTENNA CHARACTERIZATION

Item	Details
Student Name	Sunil Kumar
Roll Number	261310015
Course	SPA617 — SPASE INTRUMENTATION LABORATORY
Experiment	Antenna Characterization
Date	12/08/2026
Instructor	Prof. Mugundhan Vijayaraghavan

1. Aim

To characterize the given antenna by measuring its beam pattern in both co-polarization and cross-polarization configurations.

2. Objectives

(a) To understand the basic operating principles and construction of dipole and horn antennas.

(b) Calculate the far field distance for both (receiver and transmitter) antennas and keep separation between them greater than from the maximum far field distance of both.

(c) To study the variation of the received intensity of the incident radiation on the horn antenna with the distance between the dipole and horn antenna.

(d) To experimentally determine the directional response of the horn antenna by measuring the received signal as a function of its angular orientation for both planes in both polarizations.

(e) To obtain and compare the antenna response in the E-plane and H-plane configurations.

(f) To investigate the effect of co-axial and cross-axial polarization between the transmitting and receiving antennas on the received signal.

(g) To characterize the beam pattern of the horn antenna and identify features such as the direction of maximum response and beam width.

(h) To understand the effects of reflections, scattering, and other environmental factors on practical antenna measurements.

(i) To determine operating frequency and return loss from VSWR (Voltage Standing Wave Ratio).

(j) To measure far-field power patterns in co- and cross-polarization.

(k) To determine HPBW (Half-Power Beam Width) and estimate directivity.

3. Theory

3.1 Basics of far-field region

A far-field antenna region (or Fraunhofer region) is the space far away from an antenna where electromagnetic waves act as flat plane waves.

Far-Field distance = $\max(3\lambda, 2d^2/\lambda)$

Here, λ = wavelength, d = ((length of dipole in dipole antenna) or max(greater from aperture height or width, axial length))

3.2 Basic power concept

As an electromagnetic wave propagates through free space, its power density decreases with distance from the transmitting antenna. In the far-field region, the received power approximately follows the inverse-square law,

$$Pr \propto 1/r^2$$

Where Pr is the received power and r is the separation between the transmitting and receiving antennas.

3.1 Basic antenna concepts

An antenna is a transducer that converts electrical power into electromagnetic waves and vice versa. An antenna can be used either as a transmitting antenna or as a receiving antenna.

A dipole antenna has two equal metal arms in a straight line and is fed at the center. Its total length is approximately half a wavelength (λ/2), with a feed line connected between the two arms. It radiates most strongly in directions perpendicular to the wire and has very little radiation along the wire.

A monopole antenna is approximately half of a dipole mounted above a conducting ground plane. Its length is approximately one-quarter wavelength (λ/4). The ground acts like a mirror, making it behave like a full dipole.

- A transmitting antenna converts electrical signals into electromagnetic waves and radiates them.
- A receiving antenna converts the received electromagnetic waves into electrical signals.
- In two-way communication, the same antenna can be used for both transmission and reception.

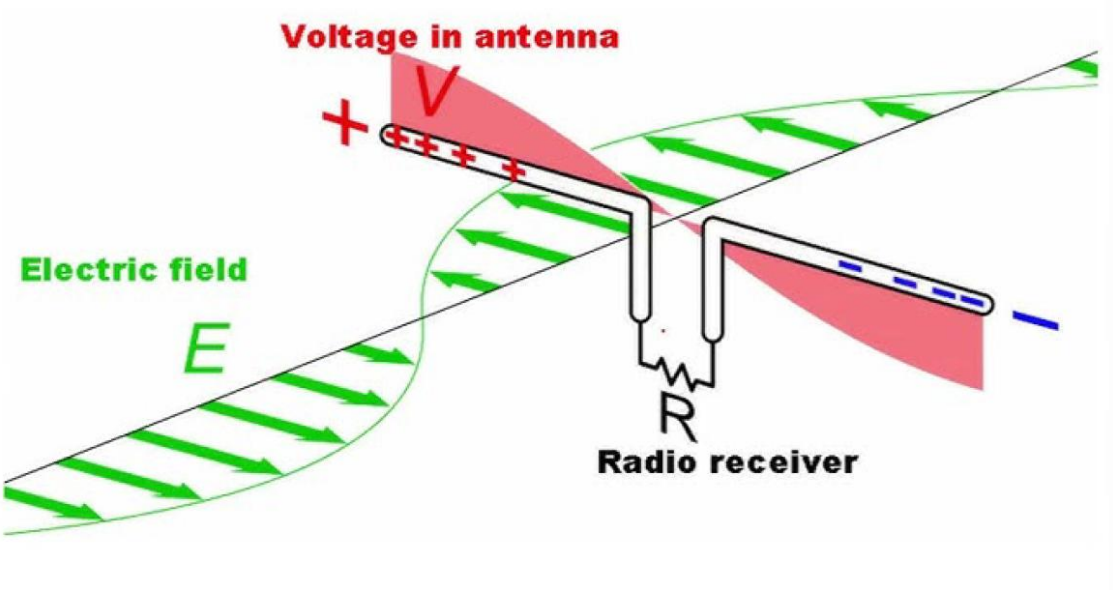


Figure 1. Half-wave dipole antenna as a receiver (image credit: Wikipedia).

3.2 Antenna beam pattern

The energy radiated by an antenna is represented by the radiation pattern of the antenna. The radiation pattern generally has a main lobe, side lobes, and a back lobe.

- The main lobe is the region where the maximum radiated energy exists. The direction of this lobe indicates the principal direction of the antenna beam.
- The other parts of the pattern where radiation is distributed sideways are known as side lobes or minor lobes.
- The lobe opposite to the main lobe is known as the back lobe.

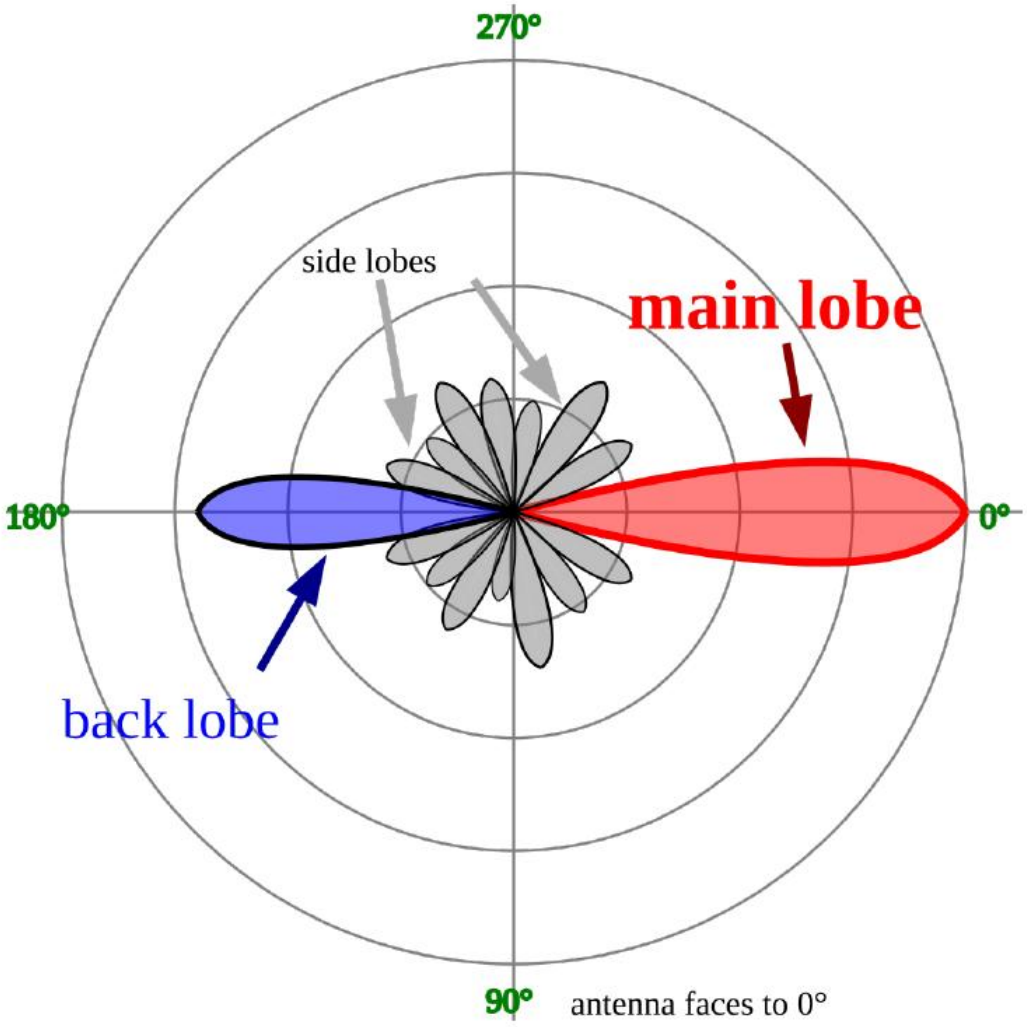


Figure 2. Antenna lobes (image credit: Wikipedia).

E-plane and H-plane angular measurements are used to determine the radiation beam pattern of an antenna. The E-plane is the plane containing the electric-field vector and the propagation vector, while the H-plane is the plane containing the magnetic-field vector and the propagation vector. The E-plane and H-plane are perpendicular to each other.

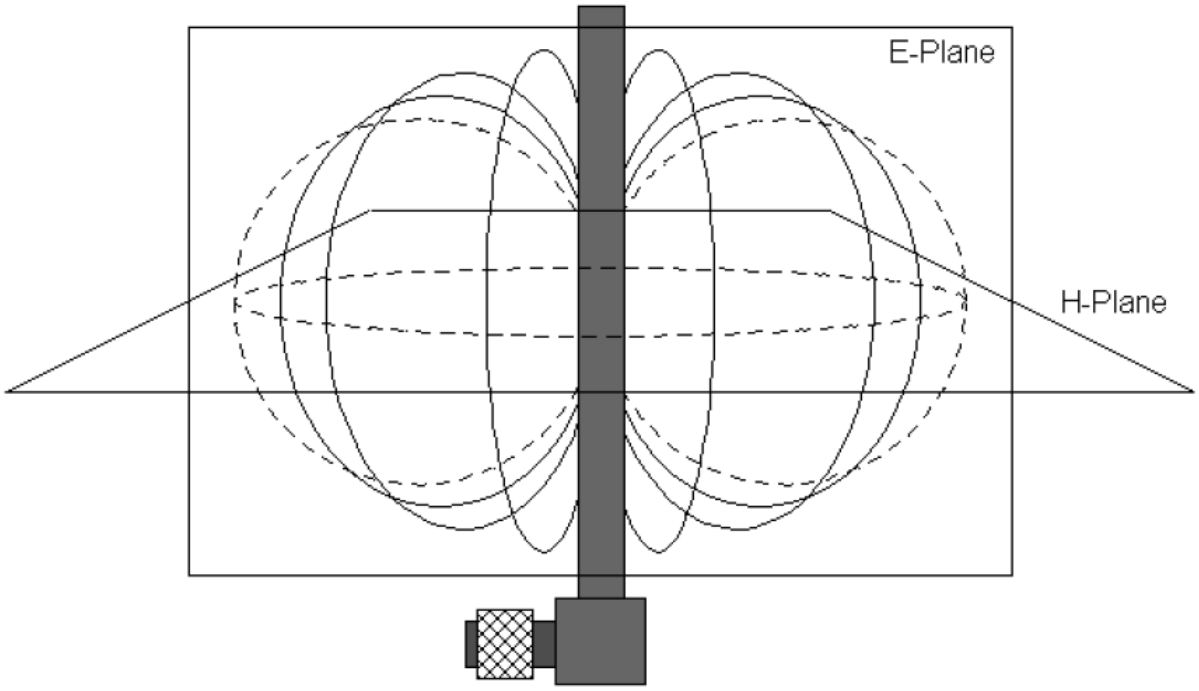


Figure – 3 E-Plane and H-Plane

3.3 Co-Polarization and Cross-Polarization

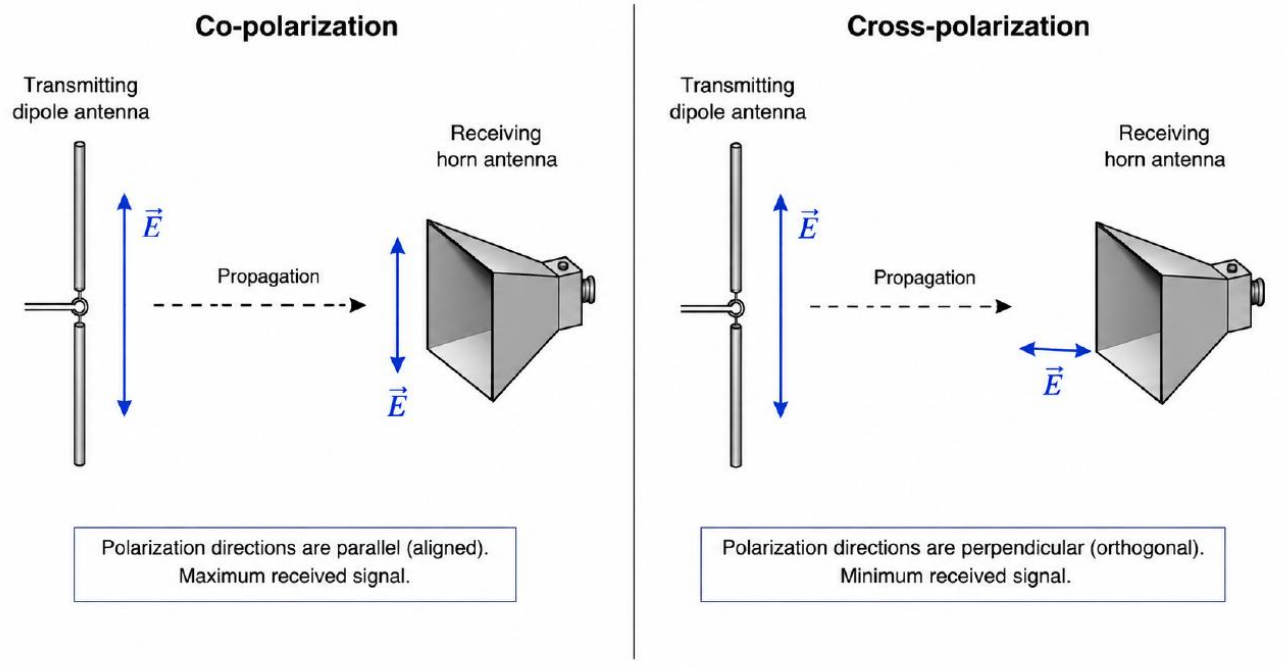


Figure – 4 Co and Cross – Polarization configuration

3.3 Reflection coefficient, VSWR, and return loss

When the antenna impedance ZL differs from the characteristic impedance Z0 of the transmission line, a part of the incident signal is reflected. The voltage reflection coefficient at the antenna terminals is

$$\Gamma = V_{reflected} / V_{incident} = (ZL - Z0) / (ZL + Z0)$$

The magnitude $|\Gamma|$ indicates the degree of impedance mismatch. For a perfectly matched antenna, ZL = Z0 and $|\Gamma| = 0$.

$$VSWR = (1 + |\Gamma|) / (1 - |\Gamma|)$$

The return loss is defined as

$$RL = -20 \log_{10} |\Gamma| \ \ dB$$

A well-matched antenna has a VSWR close to unity, a small reflection coefficient, and a large return-loss value. The resonant or operating frequency is generally identified near the minimum of the VSWR curve.

3.4 Half-power beam width

The half-power beam width (HPBW) is the angular width of the main lobe between the two points where the received power drops to half of its maximum value, corresponding to the −3 dB points.

- A small HPBW indicates a narrow, focused beam.
- A large HPBW indicates a broad beam.

3.5 Directivity

Directivity is a measure of how concentrated the radiation is in the direction of the main lobe compared with an isotropic antenna. The directivity is related to the beam solid angle ΩA as

$$D = 4\pi / \Omega_A$$

For practical antennas, the beam solid angle can be approximated using the half-power beam widths in the E-plane (θ_E) and H-plane (θ_H):

$$D \approx 4\pi / (\theta_E \, \theta_H)$$

Where θE and θH are expressed in radians.

An antenna converts electrical power into electromagnetic waves and vice versa. Its radiation pattern contains a main lobe and may contain side/back lobes. The E-plane contains the electric-field and propagation vectors; the H-plane contains the magnetic-field and propagation vectors.

4. Components

Monopole/dipole antennas; RF signal generator; DSO; spectrum analyser; RF Tee; 50 Ω termination; antenna stand; coaxial cables and RF adapters.

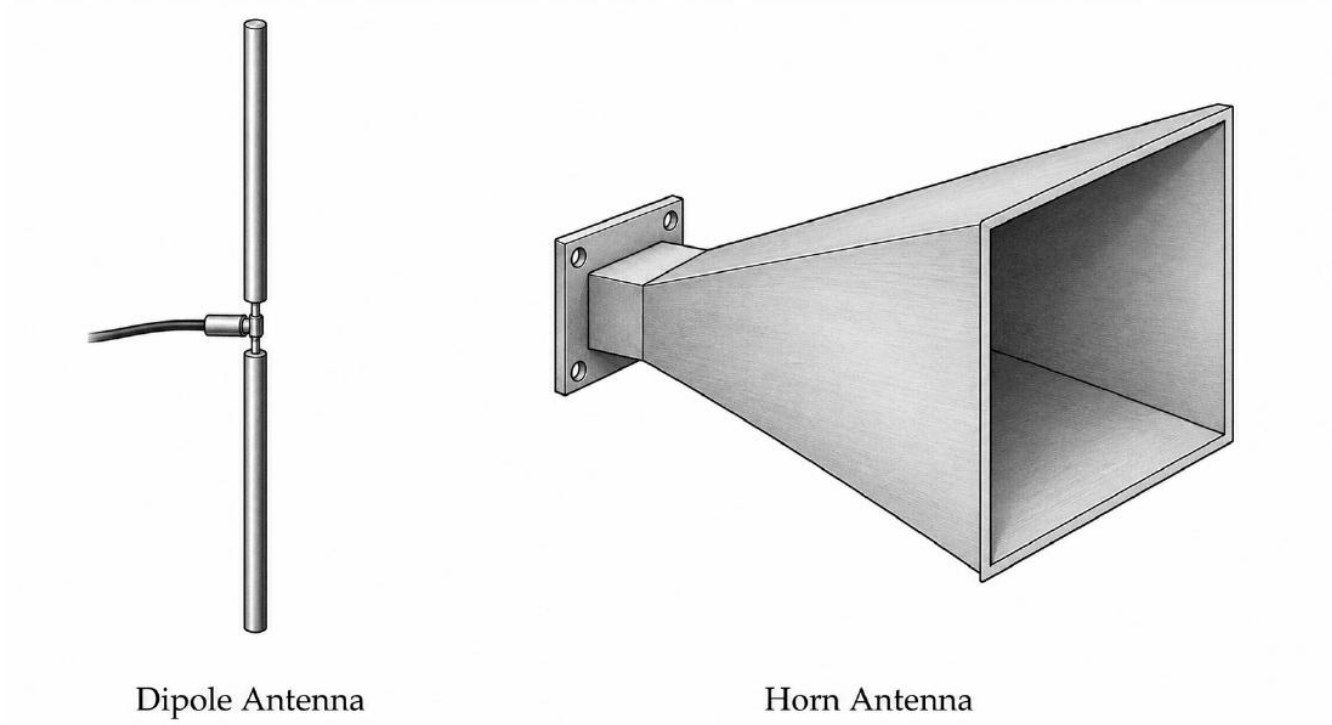


Figure – 5 Dipole and Horn antennas

5. Instrument Settings and Precautions

(a) Keep generator amplitude constant during sweeps.

(b) Set oscilloscope input to 1 MΩ for Tee measurement.

(c) Use an input signal level that is clearly measurable but does not overdrive the oscilloscope or spectrum analyser.

(c) Maintain antenna height/alignment.

(d) Keep frequency and power fixed for distance/angular measurements.

(e) Minimize nearby reflecting objects.

6. Experimental Setup

The experimental setup consisted of a dipole antenna connected to a signal generator and a horn antenna connected to a spectrum analyzer. For the distance-dependence measurement, the separation d between the transmitting dipole and receiving horn was varied while maintaining their relative alignment. For the beam pattern measurement, the separation was kept fixed and the horn antenna was rotated about its vertical axis through an angle θ. The received power at each distance or angular position was recorded using the spectrum analyzer.

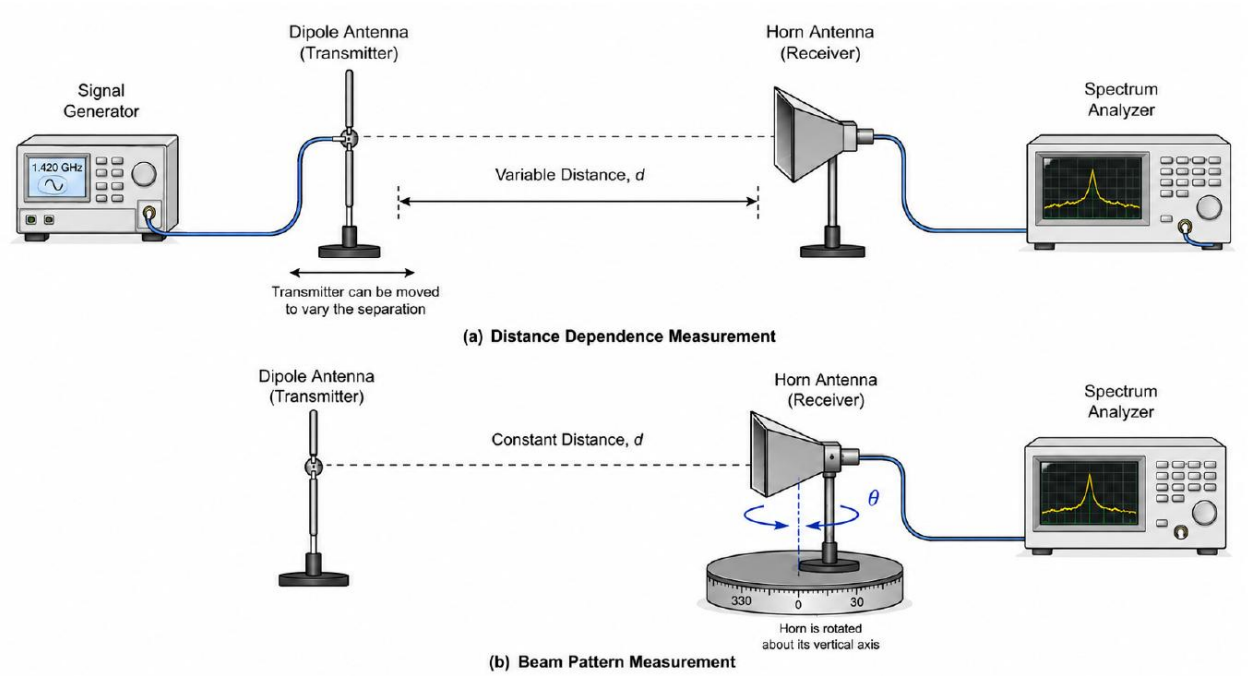


Figure – 6 Experimental setup for distance-dependence and beam-pattern measurements

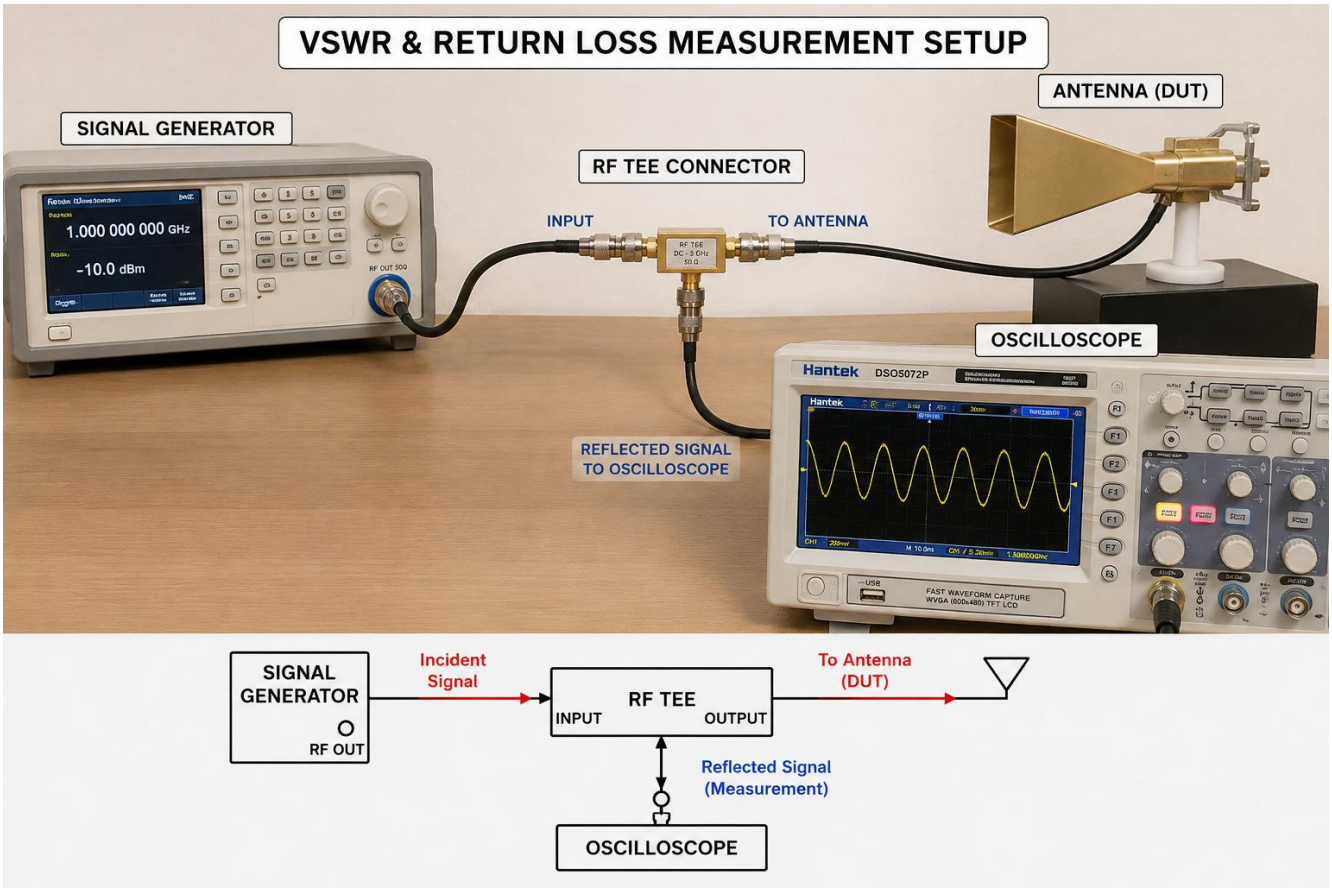


Image -1 Experimental Setup for VSWR and return-loss measurements

For the VSWR and return-loss measurement, connect the signal generator, RF Tee connector, antenna (DUT), and oscilloscope as shown in Image -1. The oscilloscope measures the variation produced by the incident and reflected signals at the Tee junction.

7. Experimental Procedure

Part A — VSWR

Connect generator, Tee, DUT and oscilloscope. Use 1 MΩ input, keep generator amplitude constant, sweep around selected frequencies, record Vmax and Vmin, calculate VSWR, $|\Gamma|$ and return loss, plot VSWR/RL and identify minimum VSWR and VSWR≤2 range.

Part B — Distance

Align transmitting and receiving antennas with same polarization and height. Keep frequency/power fixed, vary separation, record received power, and plot Pr versus R (spacing) and Pr versus $20\log_{10}R$.

Part C — Co-polarization

Keep separation ≥far-field distance. Measure E-plane and H-plane for received power from 0^0 to 360° and plot polar patterns at interval of 30^0.

Part D — Cross-polarization

Make the antenna electric fields perpendicular. Keep all other settings the same and measure E- and H-plane patterns for 0^0 to 360° and plot polar patterns at interval of 30^0.

Part E — HPBW/directivity

Find the two −3 dB points of each co-polarized main lobe, calculate θ_E and θ_H, convert to radians, and calculate D (directivity).

7. Part A — Far-Field Table

Antenna type	Maximum dimension of horn (cm)	Wavelength (cm)	Far-Field Distance (cm)
Horn	35	12.13	201.98 cm
Dipole	16	32	96 cm

7. Part B — Distance vs Received Power Table

Length (cm)	Power (dBm)	Received Power (dBm)	20 $\log_{10}$(Length)
217	20	−28.80	46.73
343	20	−26.50	50.71
418	20	−37.60	52.42
495	20	−34.55	53.89
572	20	−40.57	55.15
694	20	−46.90	56.83
726	20	−28.70	57.22
803	20	−33.63	58.09

7. Part C — Received power vs angle for the co-polarization for both planes

Distance (cm)	Angle (degrees)	Power (dBm) in E - Plane	Power (dBm) in H-Plane
500	0	−36.40	−45.24
500	30	−33.27	−48.37
500	60	−37.80	−53.92
500	90	−46.20	−62.62
500	120	−45.01	−55.09
500	150	−42.97	−51.50
500	180	−37.32	−52.32
500	210	−41.96	−43.93
500	240	−50.20	−48.85
500	270	−49.26	−57.14
500	300	−39.43	−52.72
500	330	−33.42	−45.95
500	360	−32.00	−45.22

7. Part D — Received power vs angle for the cross-polarization for both planes

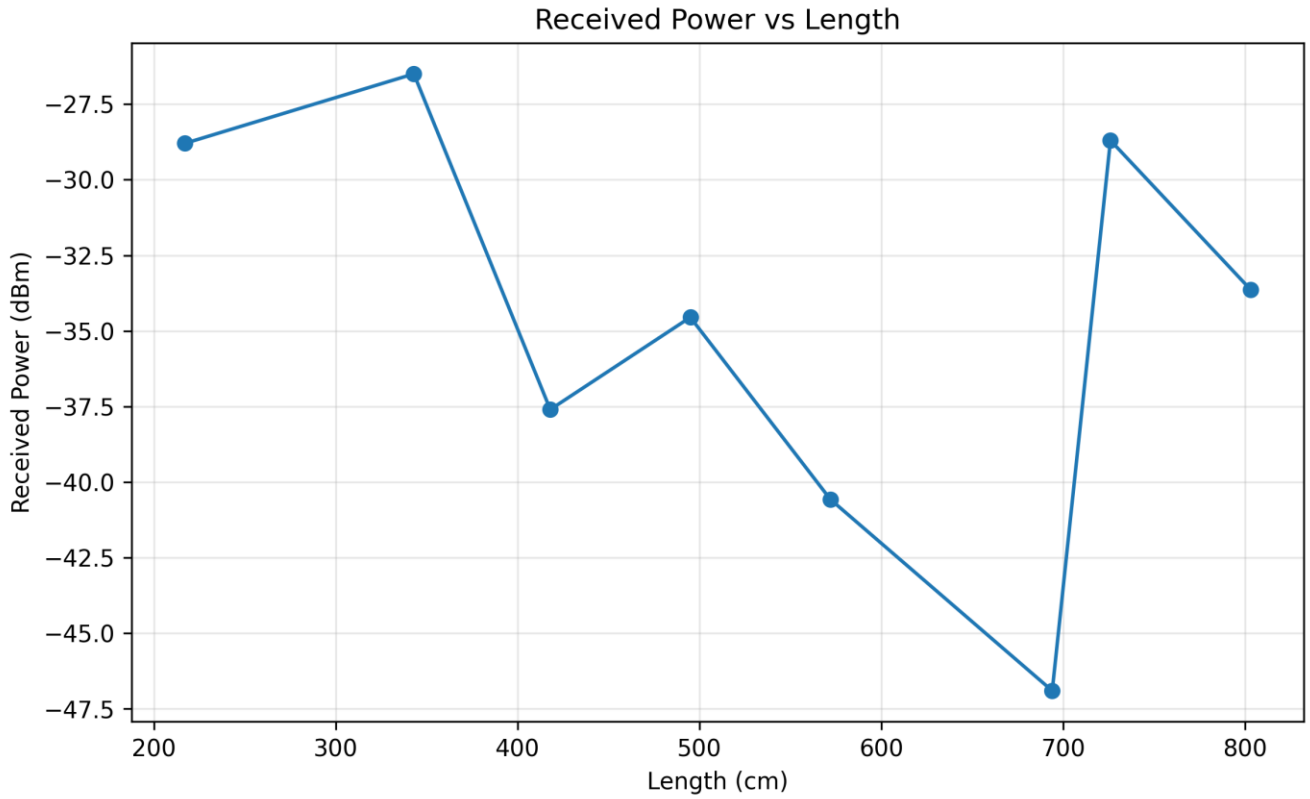
Distance (cm)	Angle (degrees)	Power (dBm) in E - Plane	Power (dBm) in H-Plane
500	0	−49.54	−41.47
500	30	−47.50	−43.32
500	60	−50.77	−40.40
500	90	−60.11	−43.09
500	120	−59.44	−42.33
500	150	−54.42	−47.71
500	180	−59.51	−41.73
500	210	−54.90	−47.40
500	240	−58.32	−59.33
500	270	−65.46	−49.34
500	300	−54.40	−44.29
500	330	−48.93	−40.96
500	360	−50.84	−40.97

7. Part E — Oscilloscope reading vs frequency table

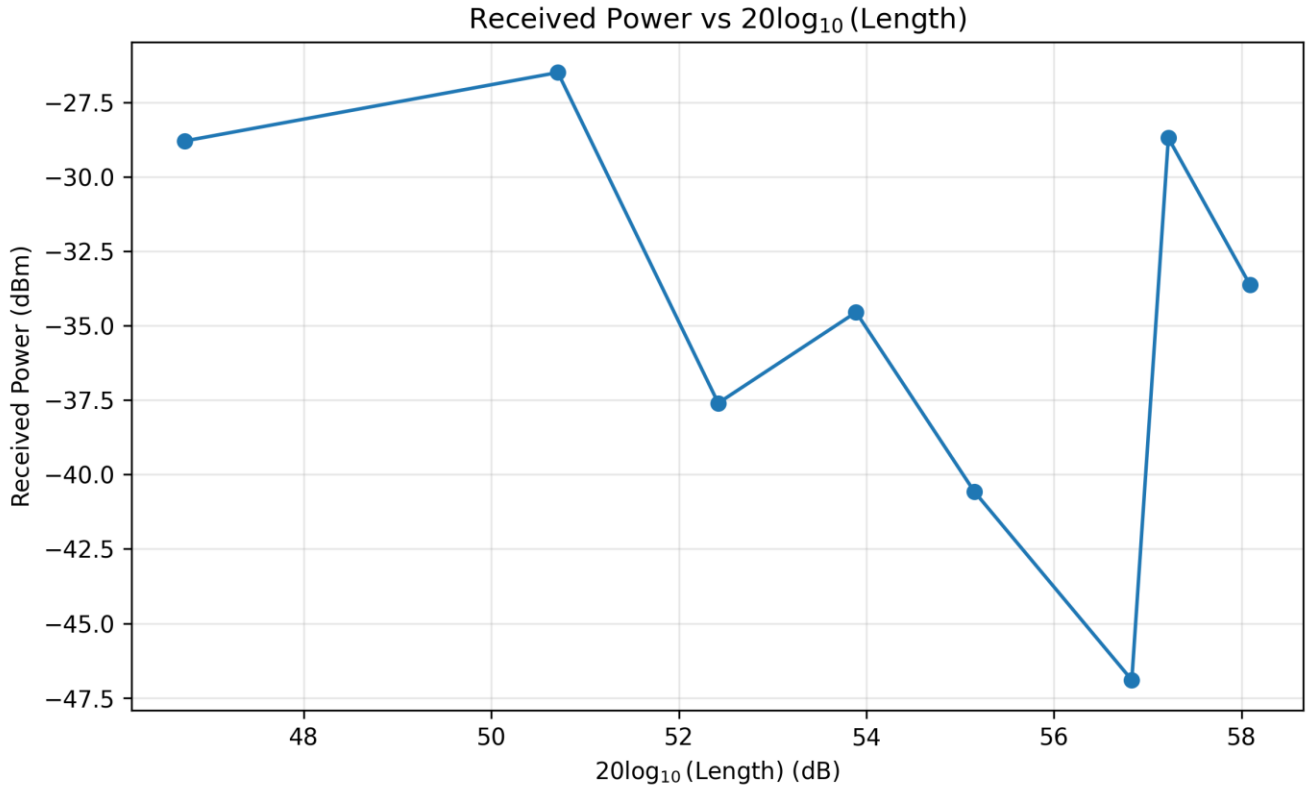
Ref. frequency or operating frequency (MHz)	Vmax of oscilloscope (volts)	Vmin of oscilloscope (mvolts)	Comments
103	1.75	700	
160.530	1.07	368.04	Corresponding frequencies are, 170.53MHz and 149 MHz

7. Part F — Graphs

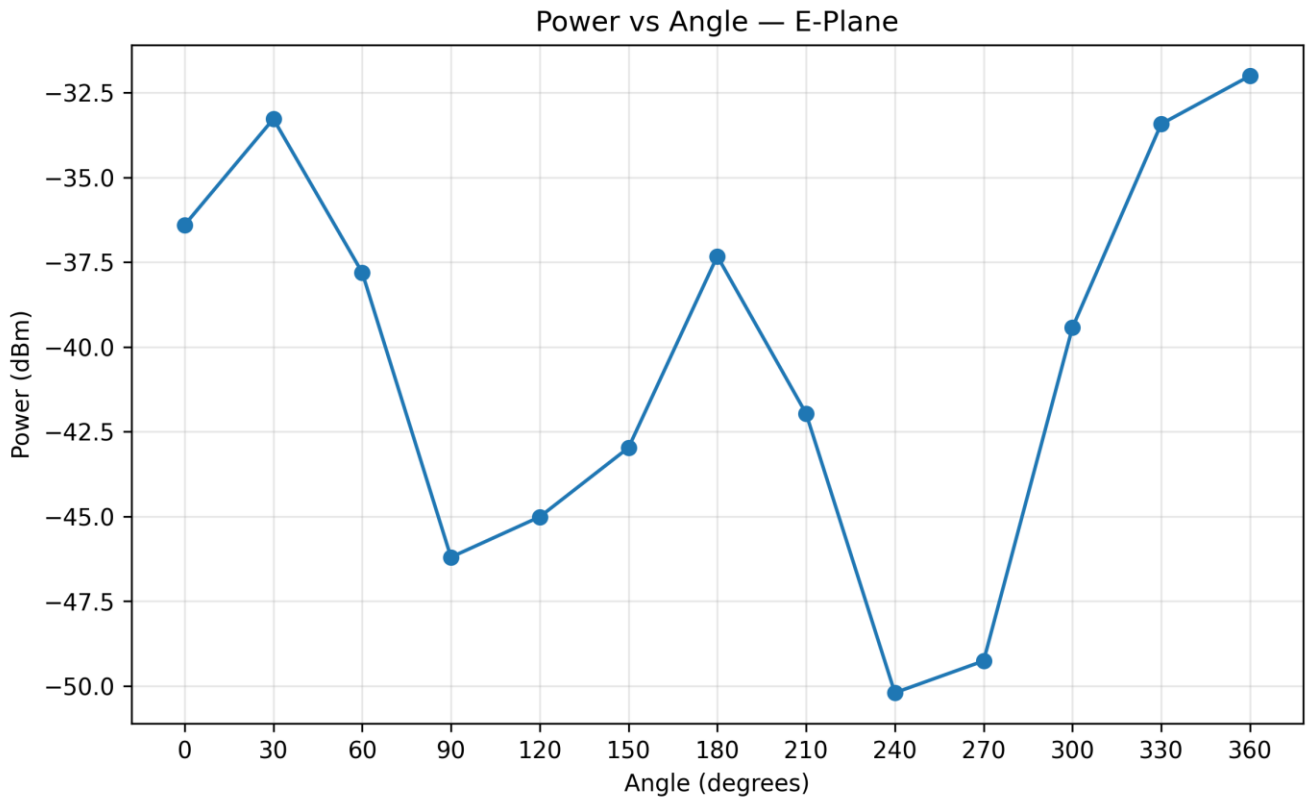
A. Received Power vs Length (spacing) -

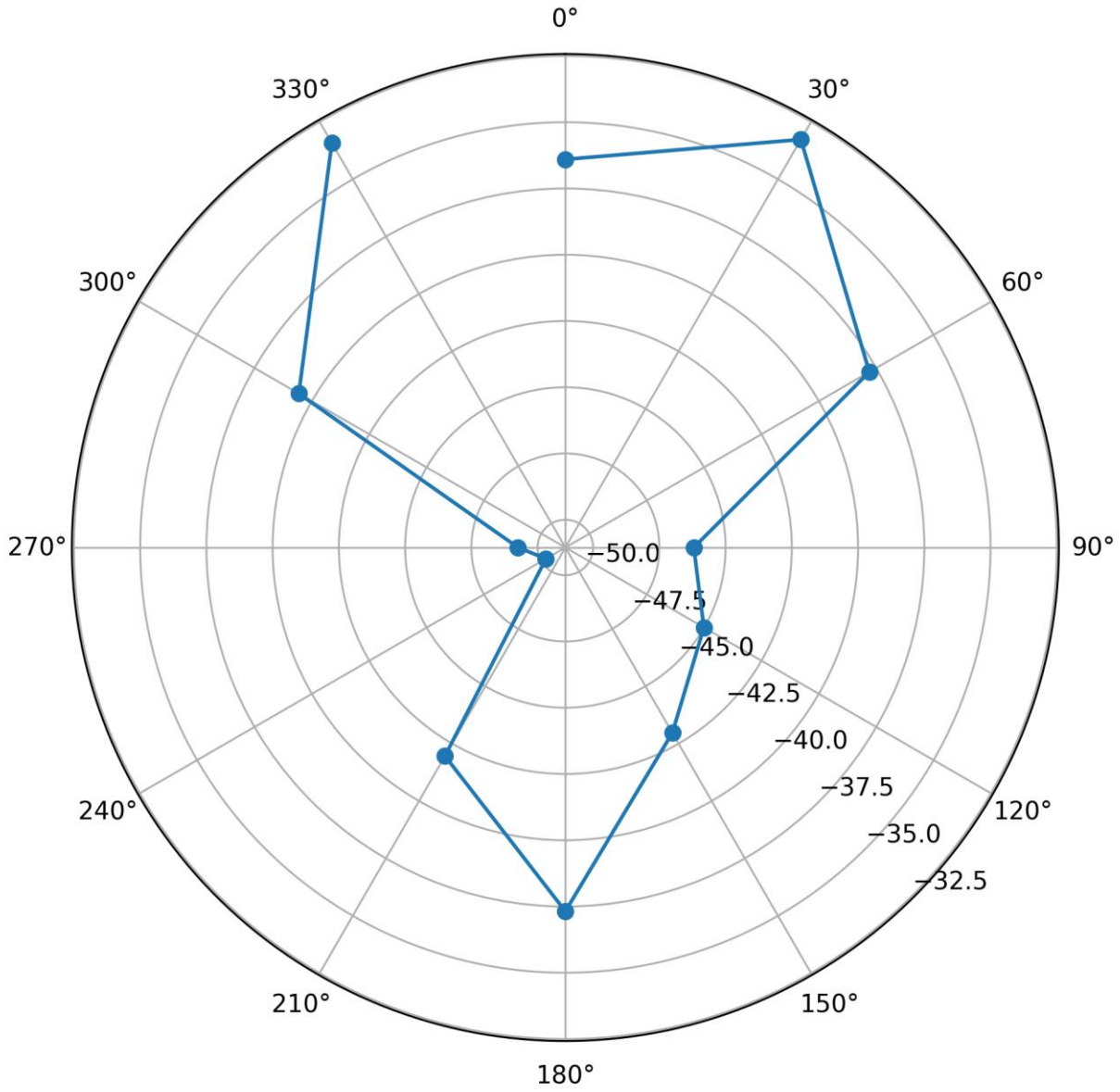


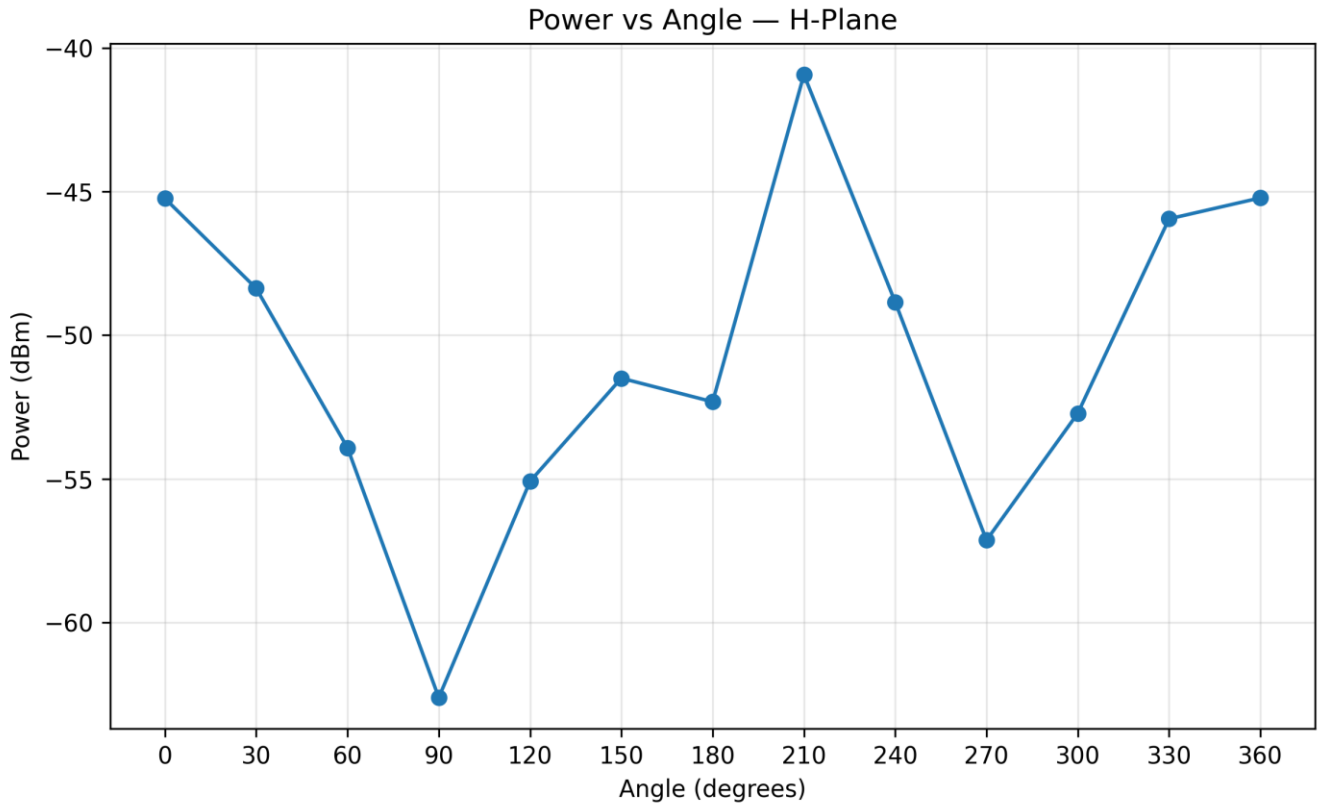
B. Received Power vs $20\log_{10}$(Length) -

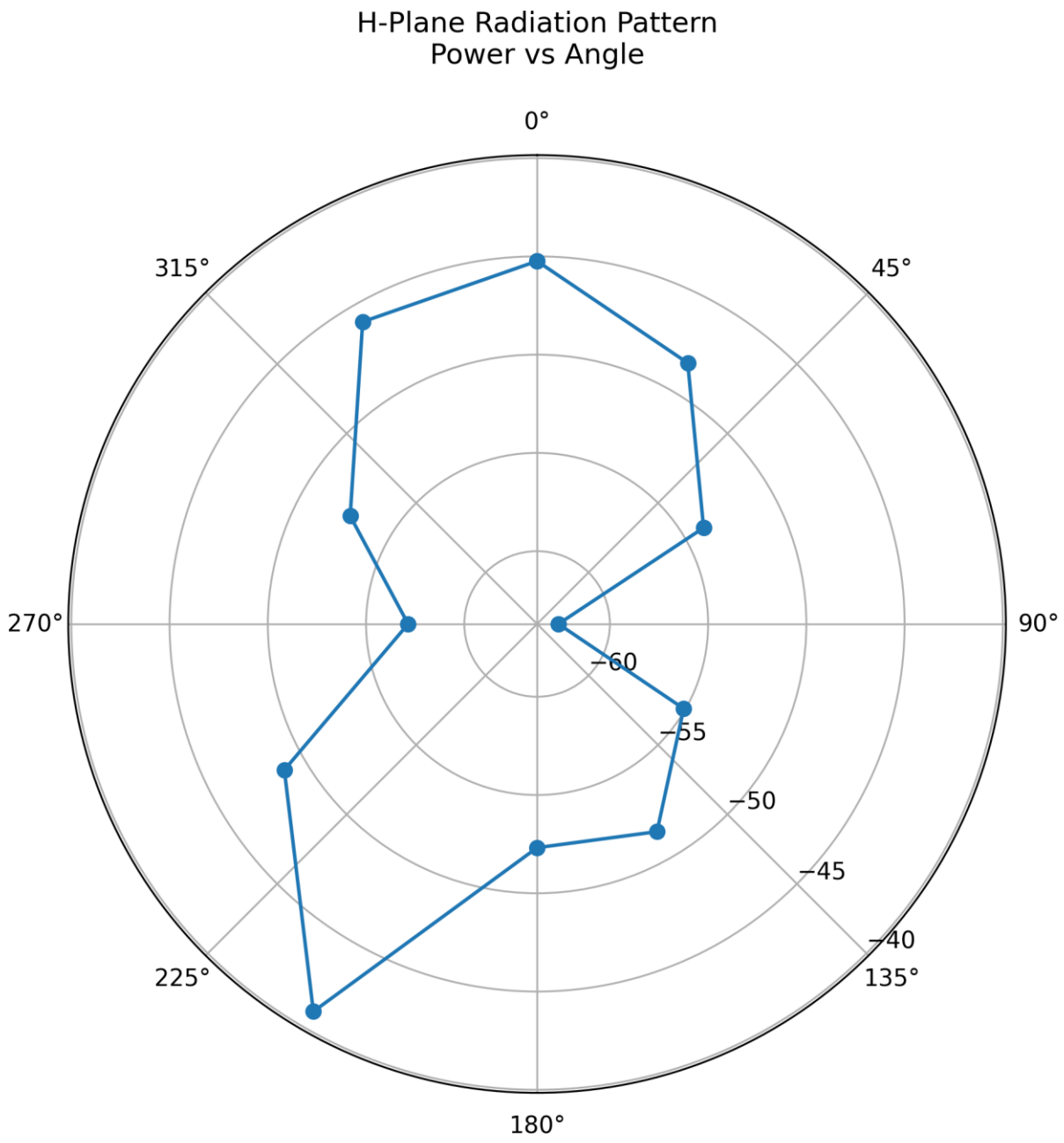


C. Received Power vs Angels in co-polarization -

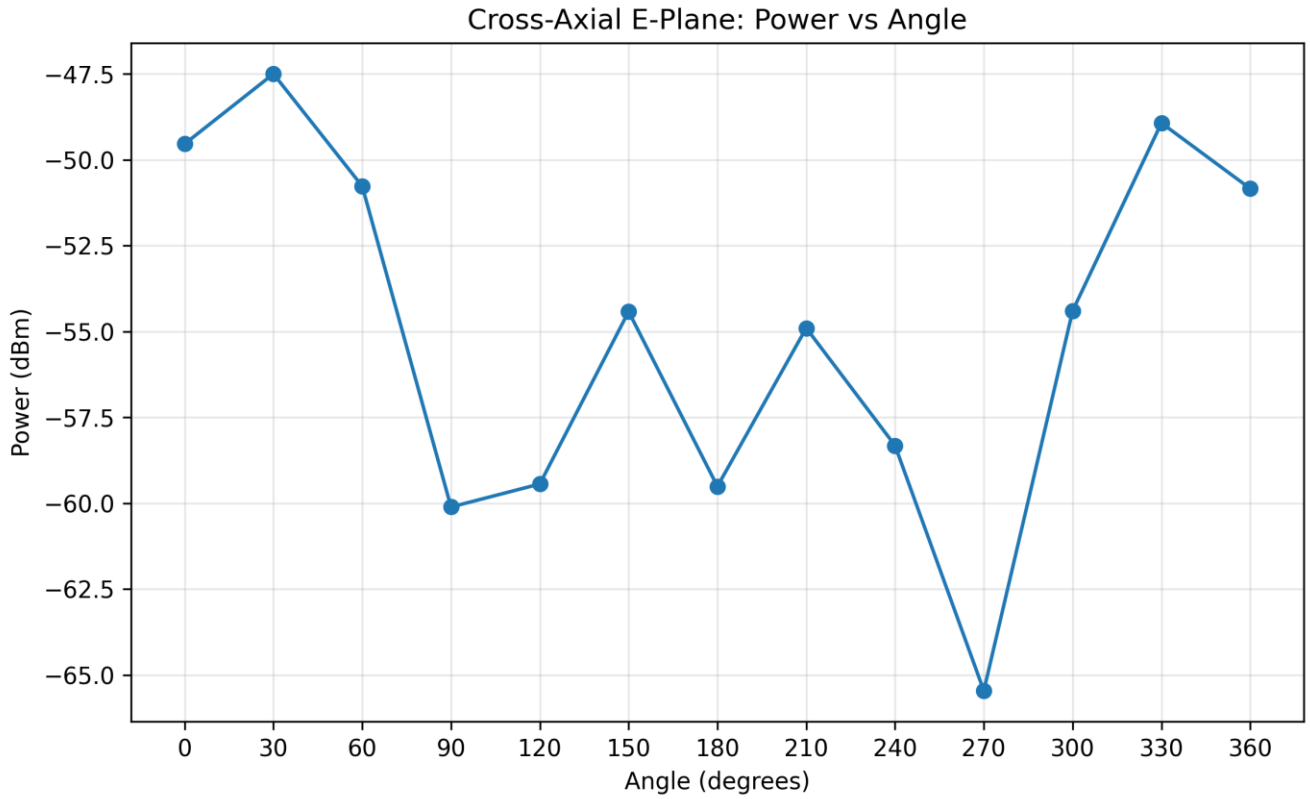

Power vs Angle — E-Plane
Power (dBm)
−32.5
−35.0
−37.5
−40.0
−42.5
−45.0
−47.5
−50.0
0
30
60
90
120
150
180
210
240
270
300
330
360
Angle (degrees)

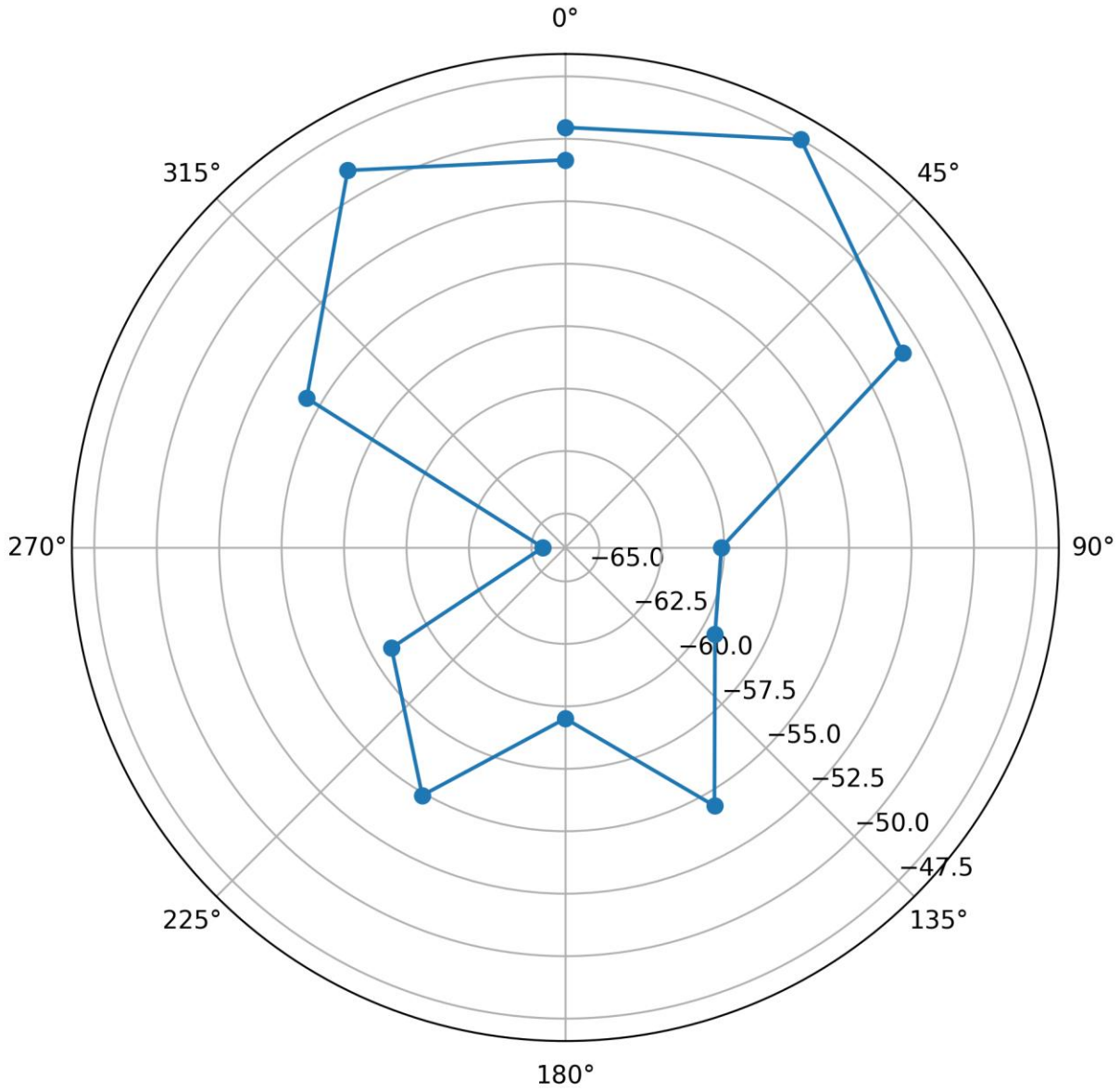

E-Plane Radiation Pattern
0°–330° measured data (360° not duplicated)
0°
30°
60°
90°
120°
150°
180°
210°
240°
270°
300°
330°
−50.0
−47.5
−45.0
−42.5
−40.0
−37.5
−35.0
−32.5

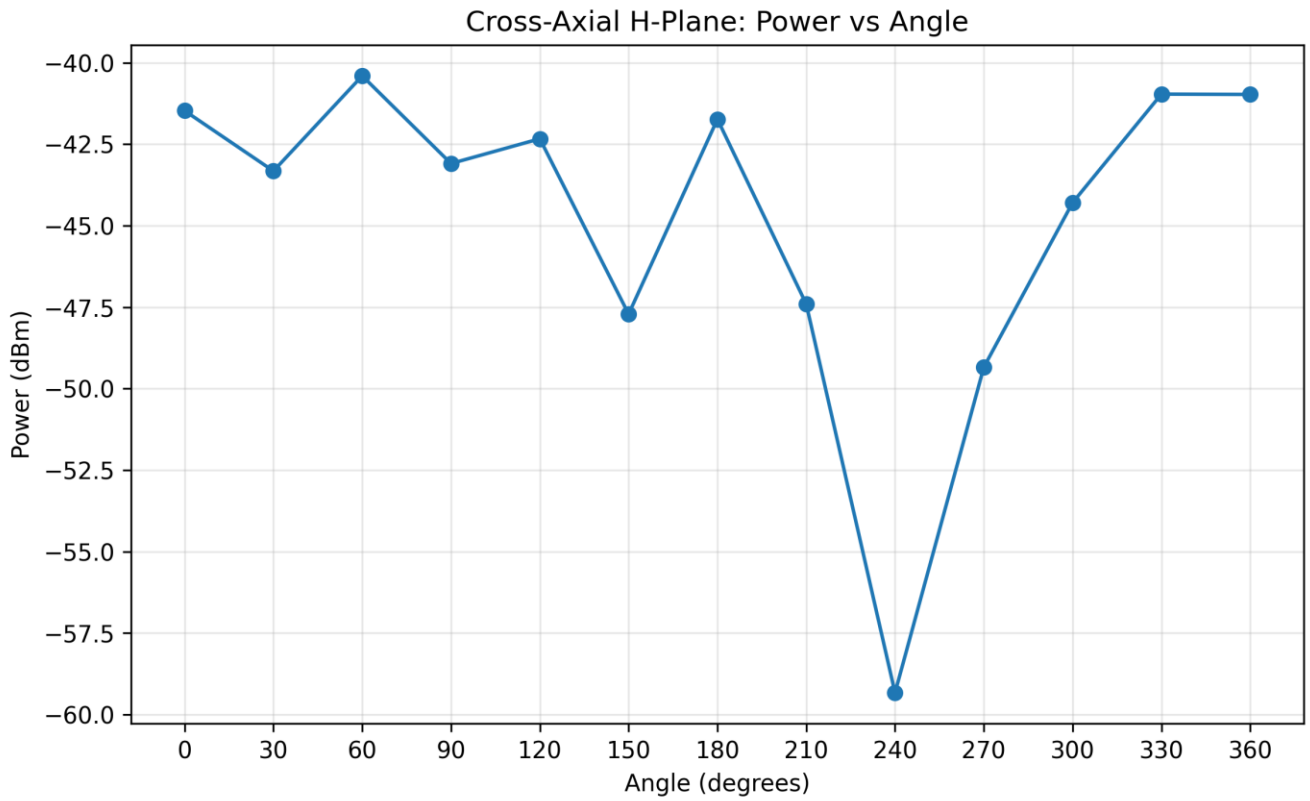

Power vs Angle — H-Plane
Power (dBm)
−40
−45
−50
−55
−60
0
30
60
90
120
150
180
210
240
270
300
330
360
Angle (degrees)

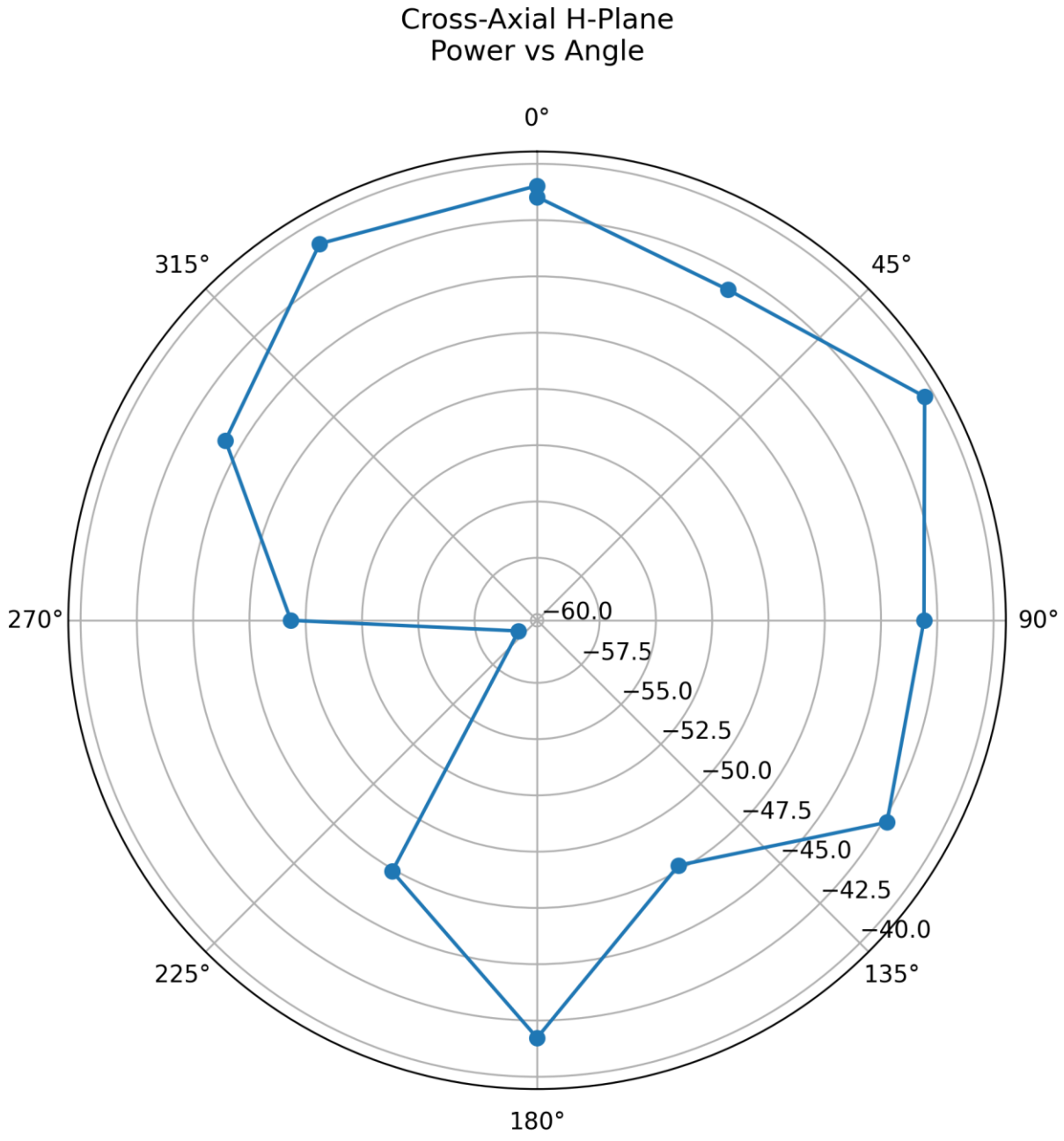


C. Received Power vs Angels in cross-polarization –


Cross-Axial E-Plane: Power vs Angle
Power (dBm)
−47.5
−50.0
−52.5
−55.0
−57.5
−60.0
−62.5
−65.0
0
30
60
90
120
150
180
210
240
270
300
330
360
Angle (degrees)


Cross-Axial E-Plane
Power vs Angle
0°
45°
90°
135°
180°
225°
270°
315°
−65.0
−62.5
−60.0
−57.5
−55.0
−52.5
−50.0
−47.5


Cross-Axial H-Plane: Power vs Angle
Power (dBm)
−40.0
−42.5
−45.0
−47.5
−50.0
−52.5
−55.0
−57.5
−60.0
0
30
60
90
120
150
180
210
240
270
300
330
360
Angle (degrees)



7.1 Sample calculation

A. Operating Frequency Calculations –

we got pole length of horn antenna is equals to 4.5 cm

So, $\lambda/4 = 4.5$ cm

Then $\lambda = 18$ cm

Then frequency = $c/\lambda = 3*10^8/18$ cm = 1.67 GHz

The actual value was 1.42 GHz.

B. Far-Field Calculations –

Far-Field distance = $\max(3\lambda, 2d^2/\lambda)$

For horn antenna d = 35 cm and λ = 12.13 cm

So, Far-Field distance = $\max(3*12.13, 2*35^2/12.13)$ = 201.98 cm

C. HPBW and Directivity Calculations –

The maximum received power in the E-plane was measured to be $P_{\max,E} = -30.72$ dBm.
The half-power level corresponds to a decrease of approximately 3 dB from the maximum received power. Therefore,

$P_{-3\text{dB},E} = P_{\max,E} - 3 = -32.00 - 3 = -35.00$ dBm

Which is coming in between 0^0 and 30^0 as −36.40 dBm and −33.27 dBm.

Using linear interpolation,
$\theta_{E,L} = 0 + (30 - 0)*(-35 + 36.40)/(-33.27 + 36.40) \approx 13.42$ degrees

$\theta_{E,L} \approx 13.42^\circ$

The second −3 dB crossing lies between 300◦ and 330◦ as -39.43 dBm and -33.43 dBm.

$\theta_{E,R} = 300 + (330 - 300))*(-35 + 39.43)/(-33.43 + 39.43) \approx 322.15$ degrees

$\theta_{E,R} \approx 360 - 322.15 = 37.85$degrees

Hence, the E-plane half-power beam width is HPBW$E = \theta_{E,R} - \theta_{E,L}$

HPBW$E \approx 37.85 - 13.42 = 24.43$ degrees $= 24.43 * \pi/180 \approx 0.43$ radians

The maximum received power in the H-plane was measured to be $P_{\max,H} = -43.93$ dBm.
The half-power level corresponds to a decrease of approximately 3 dB from the maximum received power. Therefore,

$P_{-3\text{dB},H} = P_{\max,H} - 3 = -43.93 - 3 = -46.93$ dBm

Which is coming in between 0^0 and 30^0 as −45.24 dBm and −48.37 dBm.

Using linear interpolation,
$\theta_{H,L} = 0 + (30 - 0)*(-46.93 + 45.24)/(-48.37 + 45.24) \approx 16.20$ degrees

$\theta_{H,L} \approx 16.20^\circ$

The second −3 dB crossing lies between 300◦ and 330◦ as -52.72 dBm and -45.95 dBm.

$\theta_{H,R} = 300 + (330 - 300))*(-46.93 + 52.72)/(-45.95 + 52.72) \approx 360 - 325.66$ degrees

$\theta_{H,R} \approx 34.34$ degrees

Hence, the H-plane half-power beam width is HPBW$H = \theta_{H,R} - \theta_{H,L}$

HPBWH ≈ 34.34 – 16.20 ≈ 18.14* π/180 ≈ 0.32 radians

Directivity = 4π/(0.43*0.32) = 91.32

The calculated directivity of approximately 91.32 indicates that the antenna has a preference for certain directions, despite the presence of side and back lobes in the measured beam patterns.

D. VSWR, Reflection Coefficient and Return Loss Calculations –

Minimum VSWR = Vmax/Vmin = 1.75/0.7 = 2.5
Reflection Coefficient = (2.5-1)/(2.5+1) = 0.43
Return loss = $-20 \log_{10} |\Gamma|$ db = $-20 \log_{10} 0.43$ = -12.67 db

13. Results Summary

Quantity	Sample value
Operating frequency	1.42 GHz
Minimum VSWR	2.5
Return loss	12.67 db
E-plane HPBW	24.43°
H-plane HPBW	18.14°
Estimated directivity	91.32
Reflection Coefficient	0.43

14. Discussion

The minimum VSWR indicates the frequency of best impedance matching. A lower reflection coefficient corresponds to higher return loss. Received power decreases with antenna separation. The co-polarized pattern is stronger than the cross-polarized pattern, while beamwidth provides a measure of angular concentration. Frequency changes can alter the beam shape because the electrical size of the antenna changes with wavelength.

15. Sources of Uncertainty

- Frequency resolution
- Instrument amplitude uncertainty
- Generator drift
- Antenna alignment/polarization error
- Reflections from nearby objects (in our case wall was there)
- Distance and angular-position uncertainty
- Finite angular sampling
- Cable/connector mismatch

16. Conclusion

For the illustrative data, the antenna has minimum VSWR≈2.5 near 1.42 GHz and return loss≈12.67 dB. Received power decreases with distance, and co-polarization produces a stronger response than cross-polarization. The sample HPBWs are 24.43° and 18.14°, giving estimated directivity 91.32.

17. *My Understandings –

(a) We cannot measure beam pattern before far-field distance, as optics is very complicated. Therefore, this is the limitation of this experiment.

(b) After far-field distance beam pattern is coplanar wave fronts. Therefore, we can measure output power vs distance for both the planes.

(c) Cross-polarization and H-Plane with co-polarization has minimum output. Respectively, because cross-polarization has perpendicular receiver and source antenna and H-Plane has lateral beam pattern.

(d) We are not getting inverse r^2, power vs distance because some of wall reflected waves are also interfering with the test waves.

(e) Reflection coefficient is not coming zero because of calculated operating frequency is not exactly equals to real operating frequency.

(f) As directivity value is higher, and then it is better than an isotropic antenna.